

CS272 PA2: Courier Route and SARSA(λ)

Repository

<https://github.com/Kushagrabainsla/cs272-pa2-gym>

Environment summary

The courier starts at the depot, collects a package at the center checkpoint, and delivers it at the lower-right destination. Each movement has a 15% perpendicular slip probability. The task rewards reaching both milestones while penalizing long routes.

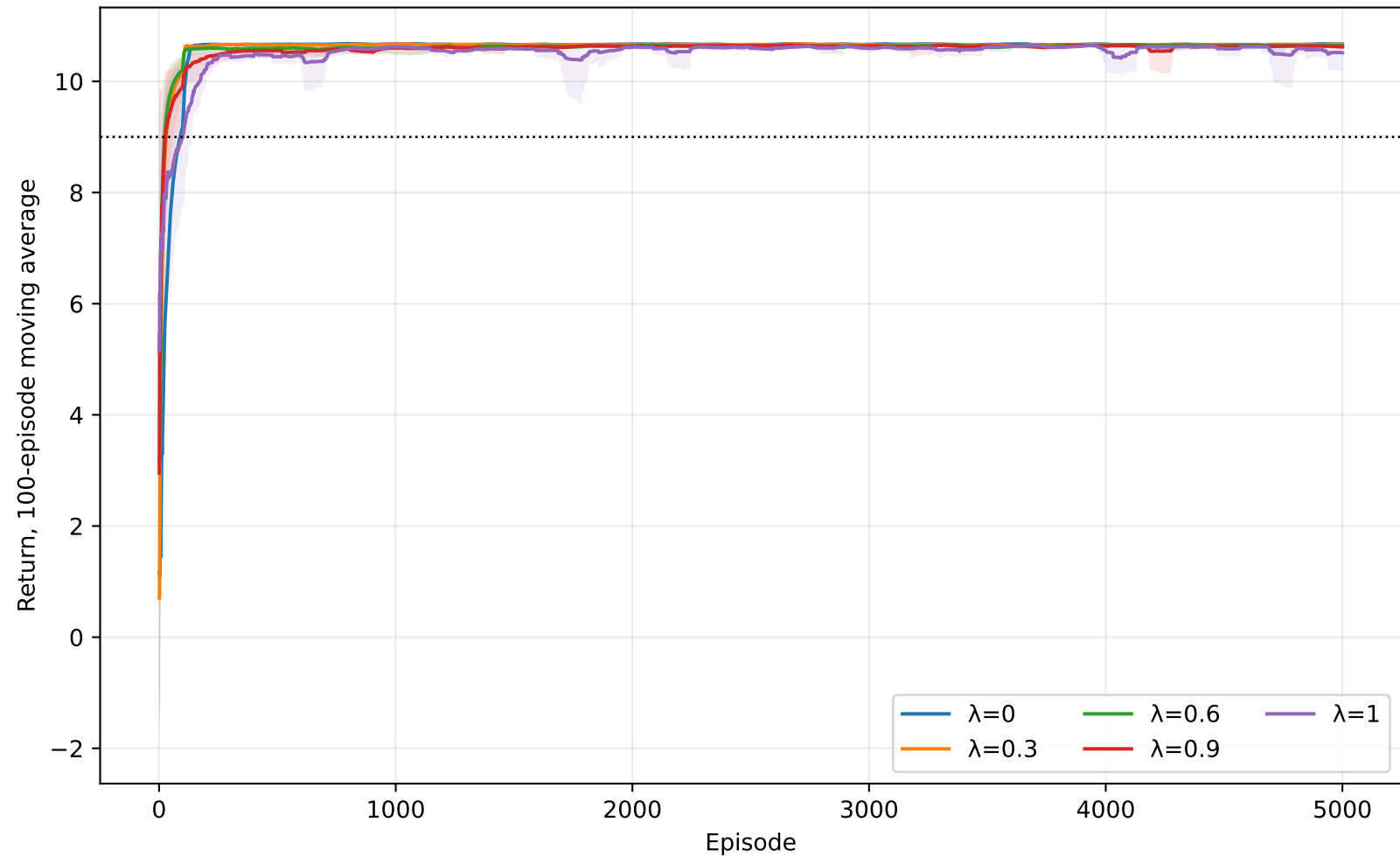
State diagram and initial ANSI rendering

Stage 0: state = row*7 + column. Reach C to enter Stage 1.
Stage 1: state = 49 + row*7 + column. Reach D to terminate.

Courier Route
Legend: C=checkpoint, D=destination, S=start, .=open, @=courier
Stage: package not collected

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@ . . . . .
. . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: -
Last reward: +0.00
```

Courier Route SARSA(λ)



Lambda sweep results

Target: smoothed mean return ≥ 9 . Each cell uses the mean over five seeded runs.

λ	First episode at target	Mean final return
0	89	10.673
0.3	30	10.667
0.6	24	10.642
0.9	27	10.621
1	99	10.516

Why λ changes the learning curve

Eligibility traces distribute a temporal-difference error backward over recently visited state-action pairs. In this environment, the checkpoint reward is about six moves from the start, and the delivery reward is about six more moves after the checkpoint. With $\lambda=0$, each reward mainly updates the immediately preceding action, so information travels backward slowly across many episodes. Larger λ values send useful reward information farther back in the same episode, which can speed learning. Very large λ can also propagate noisy slips and boundary moves farther, so the best value depends on the noise, step penalty, and learning rate.

Sample greedy episode

Reached destination: True. Return: 10.720. Each item is action:reward.

2:-0.02 2:-0.02 1:-0.02 1:-0.02 1:-0.02 2:+0.98 2:-0.02 2:-0.02 2:-0.02 1:-0.02 1:-0.02 1:-0.02 2:-0.02 1:+9.98

ANSI renderer snapshots: start, checkpoint, and terminal delivery

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Courier Route
Legend: C=checkpoint, D=destination, S=start, .=open, @=courier
Stage: package not collected
@ . . . . .
. . . . .
. . . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: -
Last reward: +0.00
```

```
Courier Route
Legend: C=checkpoint, D=destination, S=start, .=open, @=courier
Stage: package not collected
S . . . . .
. . . . .
. . . @ . . .
. . . C . . .
. . . . .
. . . . .
. . . . . D
Last action: D
Last reward: -0.02
```

```
Courier Route
Legend: C=checkpoint, D=destination, S=start, .=open, @=courier
Stage: package collected
S . . . . .
. . . . .
. . . . .
. . . . .
. . . . .
. . . . . @
Last action: R
Last reward: +9.98
```